



USN

--	--	--	--	--	--	--	--	--	--

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)

Second Semester M.Tech. (VLSI Design & Embedded Systems) (CBCS) Degree Examinations

May – June 2024

22VDE222 – EMBEDDED CONTROLLER PROGRAMMING FOR REAL-TIME SYSTEMS

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions, choosing Two full questions from Unit - I & Unit - II and
One full Question from Unit - III.

Unit – I		Marks	BT*	CO*	PO*
1.	a) What are the typical functions that a real-time computer system must perform?	10	L*2	1	3
	b) What are the main differences between a soft real-time system and a hard real-time system? Explain them briefly.	10	L2	1	3
2.	a) For an embedded system, what are the distinctive characteristics that influence the system development process, and explain briefly each one of them?	10	L2	1	3
	b) With an example, illustrate the difference between Event-triggered and Time-Triggered real-time system.	10	L3	1	3
3.	a) With the help of any two examples explain why an end-to-end protocol is required at the interface between the computer system and the controlled object?	10	L2	2	3
	b) What is the fraction of development cost/production cost in embedded systems and in plant automation systems? How does this relation influence the system design?	10	L2	2	3
Unit – II					
4.	a) Explain ARM memory organization and load store architecture.	10	L2	3	3
	b) With a neat diagram, explain ARM's visible registers and also explain in brief the ARM XPSR format.	10	L2	3	3
5.	a) With a neat block diagram, explain the three-stage pipelining architecture used in ARM processor.	10	L2	3	3
	b) With a neat block explain, how Program memory and Data memory space allocation is done in PIC18 microcontroller.	10	L2	4	3
6.	a) With examples, explain the following. (i) Immediate Addressing (ii) Direct addressing (iii) Indirect addressing Write a program to transfer the data using these three addressing modes.	10	L3	4	3
	b) With a neat block, explain how to read and write the PIC 18 Port structure.	10	L2	4	3
Unit – III					
7.	a) Explain Timer 0 in 16-bit mode for PIC 18 microcontrollers with a neat diagram. Give a brief explanation of the registers involved.	10	L2	5	3
	b) Explain in brief the registers of the PIC18 UART required while programming the controller.	10	L2	5	3
8.	a) Explain the core and peripheral interrupt sources available in the PIC18 microcontroller; also mention the ten registers that control interrupt operations.	10	L2	5	3
	b) Explain the block diagram of the PIC18F452 A/D converter and explain the registers involved in the conversion.	10	L2	5	3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
